COMPUTER FORENSICS CASE STUDY REPORT

SBT-DF204 — Case Study 1

Investigating Harassment Email Traffic With Wireshark

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Course:
SBT-DF204 Computer Forensics Case Studies

Case Title: Investigating Harassment Email Traffic With Wireshark (Nitroba University Scenario)

Analysis Workstation: Kali Linux VM

Report Date: 5 October 2026

1. Case Reference and Student Identification

Field	Value
Full name	Fasinu Temidayo Lucky
Registration number	2025/FWSD/11378
Course code	SBT-DF204 — Computer Forensics Case Studies
Assessment	Case Study 1 — Individual forensic investigation
Case title	Investigating Harassment Email Traffic With Wireshark
Scenario source	Nitroba University Harassment Scenario (digitalcorpora.org)
Analysis workstation	Kali Linux VM

This report is submitted individually. The interpretation, wording, screenshots and evidence log that follow are my own work, produced by direct analysis of the scenario-provided packet capture in Wireshark/tshark.

2. Executive Conclusion

The captured network traffic shows that, on 22 July 2008 between 02:03:43 and 02:04:24 EDT, a device on the monitored residential network (IP 192.168.15.4, Ethernet MAC 00:17:f2:e2:c0:ce, an Apple-manufactured adapter) submitted an HTTP POST to the willselfdestruct.com web service containing a message that matches — field for field — the harassing email Lily Tuckrige reported receiving. The same MAC address was observed, three minutes earlier, sending an active Gmail session cookie identifying the account **jcoachj@gmail.com**, and that account's local part is consistent with exactly one student on the Chemistry 109 roster: **Johnny Coach.**

This is strong circumstantial evidence linking a specific device and a specific Gmail account to the harassment submission, and it supports treating Johnny Coach as a person of interest warranting further investigation. It does not, by itself, prove that Johnny Coach personally typed the message. The residence's wireless network was open and shared by three students, and a MAC address identifies a device, not a human hand at the keyboard; a logged-in Gmail cookie identifies an account session, not necessarily the person who was using the browser at that moment. These limits are discussed fully in Section 7. The conclusion below is deliberately bounded to what the traffic evidence can and cannot establish.

3. Evidence Acquisition and Integrity
(Investigation Question 1)

Investigation question: What file did you acquire, and how did you preserve its integrity?

Field	Value
Evidence file name	nitroba_original.pcap
Source URL	digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Download date	5 October 2026
File size	56,180,821 bytes (54 MB)
SHA-256 (computed)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Published SHA-256 (scenario reference)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb — **matches**
MD5 (computed, secondary check)	9981827f11968773ff815e39f5458ec8 — matches published value
Working copy	working/nitroba_working.pcap — created with cp, SHA-256 verified identical to the original
Original file handling	Set to read-only (chmod 444) immediately after hashing; all analysis performed on the working copy only



Figure 3.1 — Downloading the scenario capture with wget



Figure 3.2 — Confirming the file in the evidence folder: size 56,180,821 bytes (54 MB)

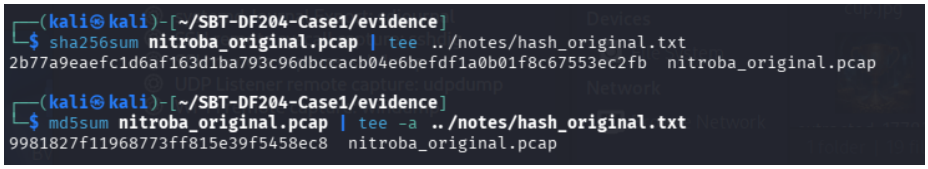


Figure 3.3 — SHA-256 and MD5 of nitroba_original.pcap, both matching the scenario's published values

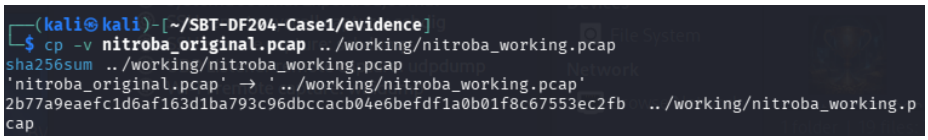


Figure 3.4 — Working copy created with cp -v, hash recorded as identical to the original

The computed SHA-256 and MD5 values match the published scenario hashes exactly, confirming the downloaded capture is authentic and has not been altered since it was published. All subsequent analysis was performed exclusively on working/nitroba_working.pcap; the original file was never opened in Wireshark or modified. Recording the source, both hash algorithms, and the working-copy procedure here allows another examiner to independently verify they are analysing the same evidence file and to reproduce every finding in this report.

Evidence reference: E01.

4. Method

The investigation proceeded in the order set out below, moving from broad identification of the relevant web service down to individual packet and byte-level verification, consistent with the lab's requirement to test and adapt filters against the actual capture rather than trust them blindly.

Step	Tool / Filter	Purpose
1	sha256sum, md5sum, cp	Acquire and hash the evidence file; create a verified working copy
2	http.request.method == "GET" && http.host == "www.willselfdestruct.com"	Identify the client and service IP addresses
3	data-text-lines contains "secure anonymous E-mail"	Independently corroborate the service responses by page content
4	ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"	Isolate the single harassment-submission POST
5	tshark -T fields -e urlencoded-form.key -e urlencoded-form.value; Follow → TCP Stream (stream 1707)	Extract and verify the POST form fields and full request/response exchange
6	tshark -T fields -e eth.src -e eth.dst; tshark -x (hex dump)	Identify and byte-verify the source/destination MAC addresses on the POST packet

Step	Tool / Filter	Purpose
7	eth.addr == <mac>; eth.src == <mac>	Establish the device's overall activity profile and first-seen packet
8	eth.src == <mac> && http.cookie contains "@"	Locate candidate identity evidence in cookies from the same device
9	grep -oE email-regex | sort | uniq -c | sort -rn	Filter 149 raw "@" hits down to genuine email addresses, removing false positives
10	Manual roster comparison	Compare the recovered account's local part against the Chemistry 109 roster
11	tshark -T fields -e frame.number -e frame.time ... (multiple filters)	Build the exact-timestamp timeline of events

Every filter listed here was run against the working copy only (nitroba_working.pcap), and every numeric or textual result quoted in Sections 5–6 was read directly from tshark/Wireshark output rather than estimated or recalled from memory.

5. Findings

5.1 Finding 2 — Which client system contacted the web service?

Filter: http.request.method == "GET" && http.host == "www.willselfdestruct.com". This returned 9 matching packets (frame numbers 82936–83654), confirming that a single client contacted the web service on 22 July 2008 between 02:03:43 and 02:04:24 EDT.

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'http.request.method == "GET" && http.host == "www.wills
elfdestruct.com"' 2>/dev/null | head -n 15
82936 15156.730593 192.168.15.4 → 69.25.94.22  HTTP 596 GET /secure/submit HTTP/1.1
82985 15157.030671 192.168.15.4 → 69.25.94.22  HTTP 355 GET /images/spacer.gif HTTP/1.1
83025 15157.238976 192.168.15.4 → 69.25.94.22  HTTP 356 GET /images/sm-logo.gif HTTP/1.1
83037 15157.279305 192.168.15.4 → 69.25.94.22  HTTP 361 GET /images/warning-home.gif HTTP/1.1
83072 15157.435074 192.168.15.4 → 69.25.94.22  HTTP 356 GET /images/body-bk.gif HTTP/1.1
83087 15157.503801 192.168.15.4 → 69.25.94.22  HTTP 358 GET /images/bttn-send.gif HTTP/1.1
83162 15157.756987 192.168.15.4 → 69.25.94.22  HTTP 361 GET /images/bridge_small.gif HTTP/1.1
83614 15197.468887 192.168.15.4 → 69.25.94.22  HTTP 461 GET /secure/success HTTP/1.1
83654 15197.806879 192.168.15.4 → 69.25.94.22  HTTP 360 GET /images/bk-message.gif HTTP/1.1
```

Figure 5.1 — The 9 GET requests to www.willselfdestruct.com (relative timestamps)

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'http.request.method == "GET" && http.host == "www.wills
elfdestruct.com"' 2>/dev/null | wc -l
9
```

Figure 5.2 — Packet count confirmed: exactly 9 matches (wc -l)

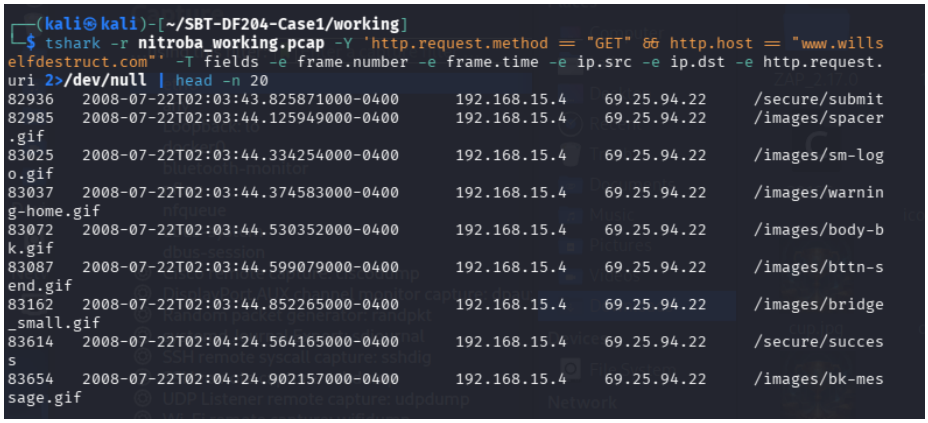


Figure 5.3 — The same 9 requests with absolute timestamps, source/destination IPs and request URIs

Field	Value	Packet #	Timestamp (EDT)
Client IP (ip.src)	192.168.15.4	82936	2008-07-22 02:03:43.825871
Service IP (ip.dst)	69.25.94.22	82936	2008-07-22 02:03:43.825871
First URI requested	/secure/submit	82936	2008-07-22 02:03:43.825871
Last URI requested	/images/bk-message.gif	83654	2008-07-22 02:04:24.902157

Corroboration: a second, independent filter searched the decoded HTTP response body for the page title text (data-text-lines contains "secure anonymous E-mail"), returning packets 82998 and 83641 — both with ip.src = 69.25.94.22 and ip.dst = 192.168.15.4, i.e. the service responding to the same client. The two response packets correspond to the initial form page and the final "success" page, matching the two-phase form-submission/display interaction described in the scenario.



Figure 5.4 — The two service responses carrying the page-title marker, corroborating the client/service pairing

Inference: 192.168.15.4 is a host on the monitored residential network (an RFC 1918 private address); 69.25.94.22 is the public-facing willselfdestruct.com web service. The client at 192.168.15.4 initiated the GET sequence during the captured window. Limitation flagged here and expanded in Section 7: an IP address alone does not identify a person — this is a shared, open wireless network.

Evidence reference: E02.

5.2 Finding 3 — What evidence links the client to the harassment message?

Filter: ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST". This returned a single matching packet, 83601, timestamped 2008-07-22 02:04:24.311700 EDT (06:04:24.311700 UTC), submitted to http://www.willselfdestruct.com/secure/submit.

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "
POST"' 2>/dev/null
83601 15197.216422 192.168.15.4 → 69.25.94.22  HTTP 719 POST /secure/submit HTTP/1.1  (application/x-www-form-urle
ncoded)
```

Figure 5.5 — The single POST packet (83601) to /secure/submit

The decoded URL-encoded form fields were extracted two independent ways — tshark field extraction and a verbose packet-detail grep — with identical results:

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -T fields -e urlencoded-form.key -e urlencoded-form.
value 2>/dev/null
to,from,subject,message,type,ttl,submit.x,submit.y      lilytuckrige@yahoo.com,,you can't find us,and you can't hi
de from us.\r\n\r\nStop teaching.\r\n\r\nStart running. ,0,30,92,26
```

Figure 5.6 — Form fields via tshark -T fields -e urlencoded-form.key/value

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -V 2>/dev/null | grep -i "form item"
    Form item: "to" = "lilytuckrige@yahoo.com"
    Form item: "from" = ""
    Form item: "subject" = "you can't find us"
    Form item: "message" = "and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running. "
    Form item: "type" = "0"
    Form item: "ttl" = "30"
    Form item: "submit.x" = "92"
    Form item: "submit.y" = "26"
```

Figure 5.7 — The same form fields via packet-detail grep, confirming consistent extraction

Form field	Value
to	lilytuckrige@yahoo.com
from	(empty — anonymised by the service)
subject	you can't find us
message	and you can't hide from us. / Stop teaching. / Start running.
type	0
ttl	30
submit.x / submit.y	92 / 26

Correlation with the reported message: the recipient address, subject line and message body in the captured POST match, field-for-field, the harassing email shown to the victim and described in the scenario. The ttl value of 30 is consistent with the scenario's description that the message was visible only briefly before the service reported it as "destroyed".

The full request/response exchange was then reconstructed with Follow → TCP Stream (tcp.stream eq 1707), which additionally reveals the client's browser identity and the server's handling of the submission:

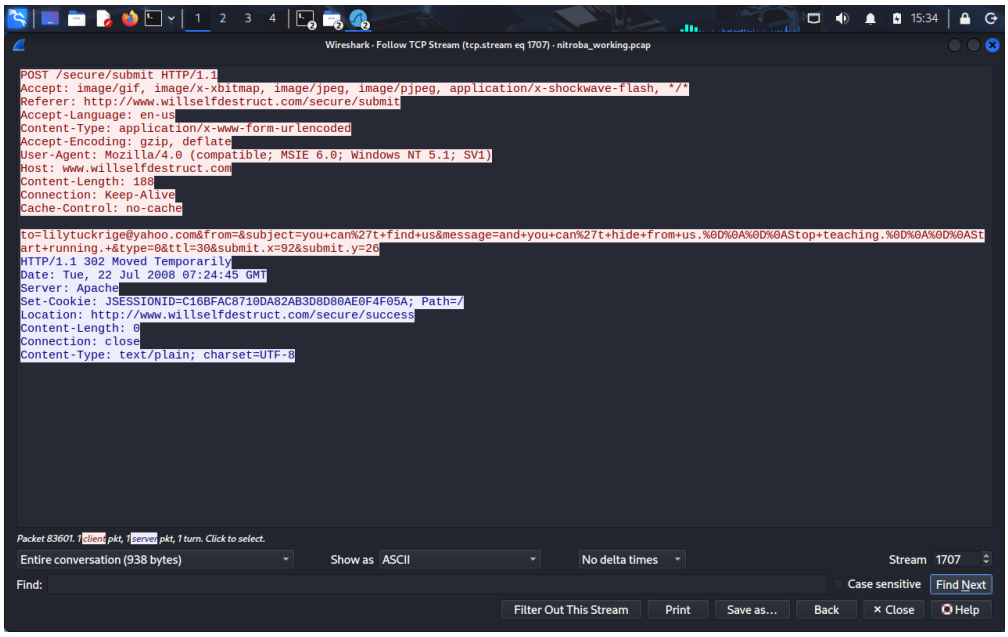


Figure 5.8 — Follow TCP Stream, tcp.stream eq 1707: full POST body, User-Agent, and the server's 302 redirect with JSESSIONID cookie to /secure/success

Field (from the stream)	Value
Client User-Agent	Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1) — i.e. Internet Explorer 6 on Windows XP
Referer	http://www.willselfdestruct.com/secure/submit
Server response	HTTP/1.1 302 Moved Temporarily → Location: /secure/success, Set-Cookie: JSESSIONID=...

Inference: the client at 192.168.15.4 submitted the harassing message through the willselfdestruct.com form-submission endpoint at 02:04:24 EDT on 22 July 2008, using a browser identifying itself as Internet Explorer 6 on Windows XP. The correlation between the captured form data and the message received by the victim is direct and unambiguous.

Limitation to flag: the POST evidence establishes what was sent and from which host on the network, but the from field has been stripped by the web service and therefore does not identify the human sender. Attribution to a person requires the additional MAC and cookie evidence in Findings 4 and 5.

Evidence reference: E03 (packet 83601), E03b (TCP stream 1707). Source MAC 00:17:f2:e2:c0:ce, destination MAC 00:1d:d9:2e:4f:60 (carried forward to Finding 4).

5.3 Finding 4 — Which device made the relevant request?

The Ethernet source and destination MAC addresses were extracted from the POST packet (83601) by field selection, then independently verified against the raw hex dump of the same packet.

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -T fields -e eth.src -e eth.dst 2>/dev/null
00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60
```

Figure 5.9 — eth.src / eth.dst extracted for frame 83601

Field	Value	Interpretation
Source MAC (eth.src)	00:17:f2:e2:c0:ce	The device that sent the POST
Destination MAC (eth.dst)	00:1d:d9:2e:4f:60	The default gateway (router) — Wireshark resolves this OUI to HonHaiPrecis (Foxconn), consistent with a consumer router

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -T fields -e eth.src -e eth.dst 2>/dev/null
00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60

┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -x 2>/dev/null | head -n 5
0000  00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00   ....O`........E.
0010  02 bd 02 ca 40 00 3f 06 c3 95 c0 a8 0f 04 45 19   ....@.?.......E.
0020  5e 16 8c cc 00 50 fc 72 02 f0 75 b8 f4 26 80 18   ^....P.r..u..&..
0030  fb 28 fa c7 00 00 01 01 08 0a 26 63 20 78 73 00   .(........&c xs.
0040  cb 95 50 4f 53 54 20 2f 73 65 63 75 72 65 2f 73   ..POST /secure/s
```

Figure 5.10 — Byte-level verification: the raw hex dump of packet 83601 shows the same MAC bytes at offset 6–11, confirming this is not a dissector artifact

```
0000   00  1d  d9  2e  4f  60  00  17  f2  e2  c0  ce  08  00  45  00
       └── dst MAC ──┘  └── src MAC ──┘  └EtherType┘
```

Manufacturer identification: the first three bytes (00:17:f2) form the OUI assigned to Apple, Inc., indicating the device is an Apple product. Wireshark's own dissector confirms this independently, resolving the address as Apple_e2:c0:ce in Figure 5.11 below (also see the frame detail in Figure 5.8's accompanying full-packet view, Section 5.2).

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'eth.addr == 00:17:f2:e2:c0:ce' 2>/dev/null | wc -l
73246

┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce' 2>/dev/null | wc -l
34582
```

Figure 5.11 — Activity-profile counts: 73,246 packets involve this MAC in either direction; 34,582 were sent by it

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce' 2>/dev/null | head -n 5
18813 9524.581107 Apple_e2:c0:ce → Broadcast    ARP 64 Who has 192.168.15.1? Tell 192.168.15.4
18815 9524.582370 192.168.15.4 → 192.168.1.254 DNS 106 Standard query 0×0000 PTR db._dns-sd._udp.0.117.168.192.in-addr.arpa
18818 9526.610956 192.168.15.4 → 192.168.1.254 DNS 78 Standard query 0×0000 A www.amazon.com
18820 9526.633433 192.168.15.4 → 72.21.210.11 TCP 82 32814 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=8 TSval=643973886 TSecr=0 SACK_PERM
18822 9526.716841 192.168.15.4 → 72.21.210.11 TCP 64 32814 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
```

Figure 5.12 — The device's first observed packets (ARP, DNS, TCP handshake to Amazon) beginning at frame 18813

Metric	Value
Total packets involving the MAC in either direction (eth.addr ==)	73,246
Packets sent by the MAC as source (eth.src ==)	34,582
First observed activity	Packet 18813 (ARP request "Who has 192.168.15.1? Tell 192.168.15.4", relative time 9524.581107 s)
IP mapping	Consistently 192.168.15.4 throughout the capture

The device's traffic includes ARP requests, DNS queries and HTTP requests spanning multiple hours, confirming it is a genuine, continuously active end-user device on the monitored network, not a spoofed one-off appearance. (Note on figures: 34,582 of the 73,246 total packets were sent by this MAC as source — both figures are reported together here because each supports a different part of the "genuinely active device" argument: the larger figure shows sustained two-way conversation involving the device, the smaller one isolates its own outbound traffic.)

Inference: the device at MAC 00:17:f2:e2:c0:ce (Apple) is the machine that sent the harassing POST. However, the MAC address alone does not identify a person. On a shared network with an open wireless access point, a MAC can in principle be spoofed or randomised, and the device may be operated by any person within physical range of the network who was using it at the time. Attribution to an individual requires the cookie/browser evidence in Finding 5. The sustained 34,582-packet, multi-hour activity profile of this MAC, however, makes a fully spoofed, one-off identity unlikely in the absence of counter-evidence.

Limitation explicitly relevant to this scenario: the residence has an open Wi-Fi router shared among three students. The MAC identifies the device, not who was using it at the time of the offence — this limitation is preserved through to the final attribution assessment (Section 7).

Evidence reference: E04a, E04b, E04c.

5.4 Finding 5 — Who is associated with the device?

With the device's MAC address identified, the capture was filtered for HTTP traffic from that device containing cookies with an "@" symbol — a common marker of an embedded email address:

```
┌──(kali㉿kali)-[~/SBT-DF204-Case1/working]
└─$ tshark -G fields 2>/dev/null | grep -iE "http.cookie" | head -n 20
F	Cookie	http.cookie	FT_STRING	http	0x0	HTTP Cookie
F	Cookie pair	http.cookie_pair	FT_STRING	http	0x0	A name/value HTTP cookie p
air
```

Figure 5.13 — Confirming the available cookie fields (http.cookie, http.cookie_pair) in this tshark build



Figure 5.14 — Raw filter eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@": 149 matching packets

The raw "contains @" filter returned 149 packets, but a sample of these (visible in Figure 5.14) are false positives — Amazon ubid-main/x-main/session tokens and other third-party tracking cookies that happen to contain an @ character but are not email addresses. These had to be filtered out before any identity conclusion could be drawn.

The matching cookie pairs were passed through a strict email-address regular expression to isolate genuine email addresses from this noise:

```
tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce' -T fields -e http.cookie_pair \
  | grep -oE "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}" \
  | sort | uniq -c | sort -rn
```

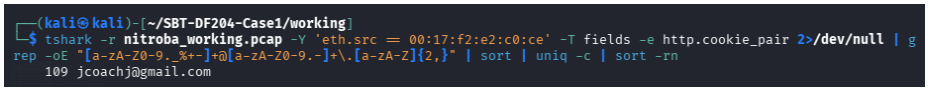


Figure 5.15 — After regex filtering, exactly one genuine email address remains: jcoachj@gmail.com, 109 occurrences

Email address	Occurrences	Source field
jcoachj@gmail.com	109	HTTP cookie (gmailchat)



Figure 5.16 — Representative gmailchat=jcoachj@gmail.com/475090 cookie values sent to Google mail servers (packets 78984, 78990, 78996)

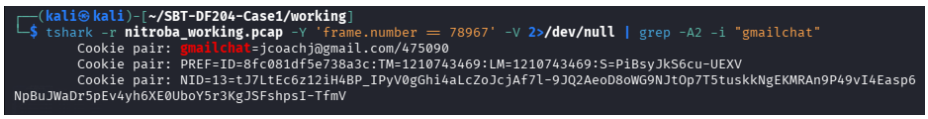


Figure 5.17 — Packet 78967 in detail: the gmailchat cookie pair alongside the Gmail session's PREF/NID cookies, confirming an active logged-in session

Packet	Timestamp (EDT)	Cookie value	Destination
78967	2008-07-22 02:01:02.113436	gmailchat=jcoachj@gmail.com/475090	74.125.19.17 (Google)
78984	2008-07-22 02:01:02.466796	gmailchat=jcoachj@gmail.com/475090	209.85.201.189 (Google)
78996	2008-07-22 02:01:02.588700	gmailchat=jcoachj@gmail.com/475090	74.125.19.17 (Google)

The destination IPs are operated by Google and host Gmail services. The timestamp of these packets — 02:01:02 EDT — is approximately 3 minutes 22 seconds before the harassment POST at 02:04:24 EDT (Finding 3), establishing that the account was actively logged in on the device immediately before the offence.

Inference: the device 00:17:f2:e2:c0:ce (an Apple machine, per Finding 4) was logged into the Gmail account jcoachj@gmail.com at the time of the harassing POST. The gmailchat cookie value embeds the account's email address directly, providing a device-to-account link.

Caveat (stronger than IP, but still not identity by itself): a cookie identifies the account logged in on that device at that moment — not necessarily the person who typed the message. MAC addresses can themselves be spoofed or randomised, and on a shared, open Wi-Fi network a MAC may belong to a device used by someone other than its usual owner.

The cookie evidence is what supplies the human-identifiable link that the IP and MAC evidence alone cannot provide, but it is a link to an account, not a sworn identity: if multiple students share the laptop, or if the account was simply left logged in, the cookie alone does not prove which human was at the keyboard at 02:04:24 EDT. This caveat governs how Finding 5 is used in the attribution assessment (Section 7) and is not resolved by any further evidence in this capture.

Evidence reference: E05.

5.5 Finding 6 — Is that person on the Chem 109 roster?

The email address recovered from the cookie evidence (jcoachj@gmail.com) was compared against the Chemistry 109 class roster provided with the case materials. The local part was decomposed into likely name components and checked against each student's surname.

Roster student	Surname	Match against jcoachj?
Amy Smith	Smith	No
Burt Greedom	Greedom	No
Tuck Gorge	Gorge	No
Ava Book	Book	No
Johnny Coach	**Coach**	**Yes**
Jeremy Ledvkin	Ledvkin	No
Nancy Colburne	Colburne	No
Tamara Perkins	Perkins	No
Esther Pringle	Pringle	No
Asar Misrad	Misrad	No
Jenny Kant	Kant	No

Only one student on the roster has the surname **Coach: Johnny Coach**. The email local part jcoachj decomposes naturally as initial j + surname coach + a disambiguating trailing j (a common pattern when the simplest form of a username is already taken).

Inference: the Gmail account jcoachj@gmail.com, whose cookies were captured on the suspect device, is plausibly associated with the student Johnny Coach on the Chemistry 109 roster. No other roster member has a name consistent with the account's local part.

Limitations: this inference relies on a naming convention, not a legal identity document. It establishes a plausible correspondence between the account and a roster member, not a proven one, and it should be presented as strong circumstantial evidence rather than direct proof of the student's

identity. It also does not prove that Johnny Coach personally typed the message — see Section 7.

Evidence reference: E06b.

5.6 Finding 7 — When did the activity occur? (summary — full timeline in Section 6)

A full chronological timeline, built directly from packet timestamps, is given in Section 6. In summary: the suspect device's Gmail session cookie was observed at 02:01:02 EDT, the first request to willselfdestruct.com at 02:03:43 EDT, the harassment POST at 02:04:24.311700 EDT, and the service's final response at 02:04:24.851431 EDT — a complete interaction window of roughly 41 seconds, with the Gmail session active at least 3 minutes 22 seconds beforehand.

Evidence reference: E06a.

5.7 Finding 8 — What conclusion can you defend? (summary — full assessment in Section 7)

The full attribution assessment, with observed facts separated from inferences and limitations, and a stated confidence level, is given in Section 7. In summary: the traffic supports treating the device at MAC 00:17:f2:e2:c0:ce, and the Gmail account jcoachj@gmail.com active on it, as strong circumstantial evidence pointing to Johnny Coach as a person of interest — not as conclusive proof of who was physically at the keyboard.

Evidence reference: E07a, E07b.

6. Timeline

All times below are reported in EDT (UTC−04:00), exactly as displayed by tshark/Wireshark on the analysis system. UTC equivalents are given in parentheses for external verification. No other time-zone conversion was applied anywhere in this report.

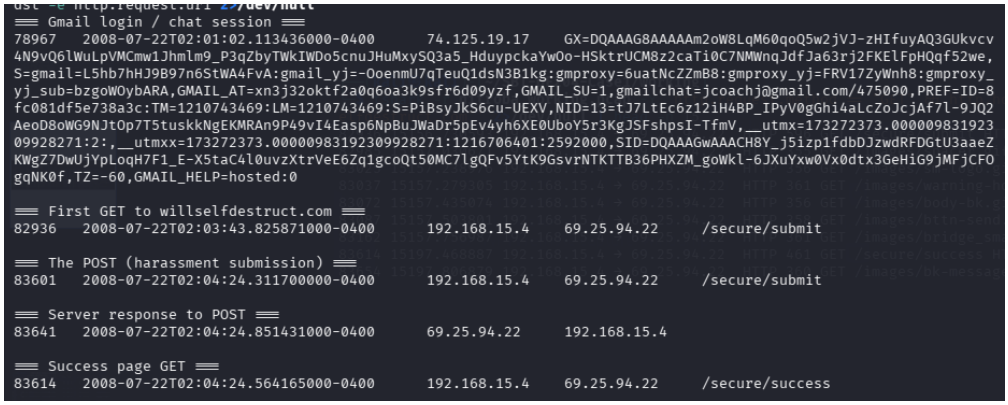


Figure 6.1 — Composite tshark timeline: Gmail login/chat session, first GET to willselfdestruct.com, the harassment POST, and the server's response

#	Event	Packet	Timestamp (EDT)	UTC (reference)
1	Suspect device sends Gmail chat cookie (gmailchat=jcoachj@gmail.com/475090) to Google	78967	2008-07-22 02:01:02.113436	2008-07-22 06:01:02 UTC
2	Client requests /secure/submit from willselfdestruct.com (first page load)	82936	2008-07-22 02:03:43.825871	2008-07-22 06:03:43 UTC
3	Service responds with the initial web form (title contains "secure anonymous E-mail")	82998	2008-07-22 02:03:44.210946	2008-07-22 06:03:44 UTC
4	Client submits the harassment form via HTTP POST	83601	2008-07-22 02:04:24.311700	2008-07-22 06:04:24 UTC
5	Service responds with the "success" page (same title marker)	83641	2008-07-22 02:04:24.851431	2008-07-22 06:04:24 UTC
6	Client requests success-page assets (/secure/success, /images/bk-message.gif)	83614, 83654	2008-07-22 02:04:24.564165 – 02:04:24.902157	2008-07-22 06:04:24 UTC

Observations:

- The suspect's device was already logged into the jcoachj@gmail.com Gmail account at least 3 minutes 22 seconds before the harassment was submitted.

- The complete interaction with willselfdestruct.com — from first page load to last asset — spans approximately 41 seconds (02:03:43 → 02:04:24 EDT).
- The harassment submission itself occurred at 02:04:24.311700 EDT (06:04:24.311700 UTC).

Evidence reference: E06a.

7. Attribution Assessment

7.1 Identity chain summary

192.168.15.4 (IP)	→	00:17:f2:e2:c0:ce (MAC)	→	jcoachj@gmail.com (Account)

Each arrow in this chain is supported by independent packet evidence (Findings 2, 4 and 5 respectively), but each arrow also carries its own limitation — summarised after the chain and detailed below — so the chain as a whole should be read as a sequence of corroborating links, not as a single proof.

7.2 Observed facts (Tier 1 — direct from packets)

- At 22 July 2008, 02:03:43–02:04:24 EDT, the host 192.168.15.4 on the monitored residential network interacted with the willselfdestruct.com web service at 69.25.94.22.
- Packet 83601 is an HTTP POST to /secure/submit containing the form fields to = lilytuckrige@yahoo.com, subject = "you can't find us", message = "and you can't hide from us. / Stop teaching. / Start running."
- The Ethernet source MAC of packet 83601 is 00:17:f2:e2:c0:ce (Apple OUI), verified independently at the raw byte level; the destination MAC is the network gateway.
- The same MAC transmitted HTTP cookie pairs containing gmailchat=jcoachj@gmail.com/475090 to Google mail servers in 109 packets, including packet 78967 at 02:01:02 EDT — 3 minutes 22 seconds before the POST.
- The Chemistry 109 roster provided with the case contains exactly one student whose surname matches the email local part: Johnny Coach.

7.3 Reasonable inferences (Tier 2 — evidence-based but inferential)

- The device at MAC 00:17:f2:e2:c0:ce is a single, consistently active Apple machine on the monitored network (73,246 packets involving it over multiple hours, 34,582 sent by it as source).
- The Gmail account jcoachj@gmail.com was logged into that device at the time of the harassment POST.
- The email local part jcoachj is consistent with the name "J. Coach", matching Johnny Coach on the Chem 109 roster.
- The harassment message submitted via the POST matches the content the victim reported receiving via willselfdestruct.com.

7.4 Limitations and alternative explanations (Tier 3)

1. Shared open wireless network. The residence has an unauthenticated Wi-Fi router shared among three students. The MAC identifies a device on the network, not the human operator at the keyboard.

2. MAC spoofing is possible. The source MAC is asserted by the packet, not cryptographically verified. However, the sustained 34,582 packets sent by this MAC over a multi-hour window make a fully spoofed, one-off identity unlikely in the absence of counter-evidence.

3. Cookie ≠ human identity. Cookies prove the account was logged into the device, not that the account owner was the one who submitted the message. Accounts can be left logged in on a shared machine, and if three students share the laptop, the cookie alone cannot distinguish which of them was at the keyboard at 02:04:24 EDT.

4. Username inference. jcoachj is a strong indicator of "J. Coach", but a username is a naming convention, not a legal identity document.

5. Subnet/segment change between the two reported incidents. The full headers from the earlier, first harassment email point to IP address 140.247.62.34 — a different address from the 192.168.15.4 host identified in this capture. Both are internal Nitroba addresses, but they sit on different subnets. This is because the monitoring sniffer was relocated to a different segment of the network (the 192.168.15.x residence subnet) by the time the second message was sent through willselfdestruct.com on 21–22 July; the two addresses are not expected to match, and this capture speaks only to the second incident, not the first.

6. Blank "from" field. willselfdestruct.com anonymised the sender's stated address, so the POST itself does not name a sender; the sender-identifying

evidence in this report comes entirely from the MAC and cookie correlation, not from the form content.

7.5 Confidence level

Statement	Confidence
The device at MAC 00:17:f2:e2:c0:ce submitted the harassment POST.	High
The Gmail account jcoachj@gmail.com was logged in on that device at the time.	Medium-high
The human operator at the keyboard was Johnny Coach.	Medium — limited by the shared network and the cookie-to-human gap

7.6 Defensible conclusion

The evidence supports a conclusion that the harassment message was submitted from the device at MAC 00:17:f2:e2:c0:ce (an Apple machine on the monitored residential network), while a Gmail session for jcoachj@gmail.com was active on that device. The email local part is consistent with the student Johnny Coach on the Chemistry 109 roster, and no other roster member provides a name match. On a shared, open wireless network, however, the captured traffic cannot by itself prove which human was at the keyboard. The finding is best characterised as strong circumstantial evidence linking a specific device and account to the harassment, supporting further investigation of the account holder as a person of interest, rather than as conclusive proof of who physically sent the message.

Evidence reference: E07a, E07b.

8. Appendix A — Evidence Log

Evidence ID	Packet / item	Finding	Why it matters	Screenshot
E01	nitroba_original.pcap + working copy	Source downloaded; SHA-256/MD5 match published values; working copy byte-identical; original set read-only	Proves evidence integrity and chain of custody for the entire investigation	1, 1a, 2, 3 (Section 3)
E02	Packets 82936–83654 (GETs to willselfdestruct.com)	Client IP 192.168.15.4 contacted service IP 69.25.94.22 on 22 Jul 2008, 02:03:43–02:04:24 EDT	Identifies the two endpoints involved in the incident; corroborated by two independent filters	6, 6b, 7, 8 (Section 5.1)
E03	Packet 83601	HTTP POST to /secure/submit with to=lilytuckrige@yahoo.com, subject="you can't find us", message="and you can't hide from us..."	Directly matches the content of the harassing message received by the victim; establishes what was sent and when	9, 11.POST_inspect, 11.POST_Grep (Section 5.2)
E03b	TCP stream 1707	Follow TCP Stream of the POST/response exchange; reveals client User-Agent (IE6/WinXP) and server 302 redirect to /secure/success	Provides complete request-response context and confirms the POST was accepted by the server	TCP_stream (Section 5.2)
E04a	Packet 83601 (eth.src field)	Source MAC 00:17:f2:e2:c0:ce identified as the device that sent the POST	Unique device identifier; stronger than IP for forensic attribution	15 (Section 5.3)
E04b	Packet 83601 (hex dump)	Raw bytes at offset 6–11 confirm source MAC 00:17:f2:e2:c0:ce at the byte level	Independent verification of the MAC at the raw packet level; defeats a "dissector artifact" challenge	12 (Section 5.3)
E04c	MAC 00:17:f2:e2:c0:ce (all traffic)	Device sent 34,582 packets and appeared in 73,246 total packets across the capture, with stable mapping to 192.168.15.4	Confirms the device is a genuine, consistently active end-user machine, not a transient or spoofed one-off	13, 14.MAC_packets_sent, 14.MAC_packets_sent_000 (Section 5.3)
E05	Packets 78967, 78984, 78996 (+106 others)	HTTP cookies from MAC 00:17:f2:e2:c0:ce contain gmailchat=jcoachj@gmail.com/475090, 109 times, sent to Google mail servers at 02:01:02 EDT	Links the device to a specific Gmail account; establishes the account was active on the device minutes before the POST	16, 17.cookies, 17.cookies_capture, 18, 19 (Section 5.4)
E06a	Packets 78967, 82936, 82998, 83601, 83614, 83641, 83654	Chronological sequence: Gmail session active at 02:01:02, first GET at 02:03:43, POST at 02:04:24, success page at 02:04:24	Establishes the order and timing of relevant events; shows the suspect account active on the device minutes before the offence	20 (Section 6)
E06b	Chemistry 109 roster vs jcoachj@gmail.com	Only one roster member has the surname "Coach" — Johnny Coach — and the local part decomposes naturally as "J. Coach"	Links the captured account to a named individual on the roster; a strong circumstantial match but not proof of identity	Roster comparison table (Section 5.5) — no screenshot; based on roster text supplied with the case
E07a	Combined chain (packets 78967, 82936, 83601, 83641)	Full evidence chain: Gmail session → willselfdestruct.com GET → POST with harassment content → server response	Summarises the complete evidence chain supporting the attribution	Section 7.1 identity-chain diagram

E07b	Attribution assessment	Direct observations, reasonable inferences, limitations, confidence level and defensible conclusion	Fulfils the marking guide's requirement for a careful, cautious, evidence-bounded conclusion	This report, Section 7

Appendix B — Screenshot Reference

#	Screenshot	Evidence Shown
1	1.pcap_file.png	wget download of the scenario capture
2	1a.pcap_file_evidence.png	Evidence folder listing confirming file size
3	2.pcap_hash.png	SHA-256 and MD5 of the original file
4	3.working_pcap.png	Working copy creation and hash verification
5	4.evidence_1.png	Evidence log (evidence_log.md) as recorded contemporaneously
6	5.tshark_reading.png	Initial tshark read-through of the working copy
7	6.web_request.png / 6b.web_request_no.png	GET requests to willselfdestruct.com and packet count
8	7.first_GET.png	Full field extraction of the 9 GET requests with absolute timestamps
9	8.page_title.png	Service responses carrying the page-title marker
10	9.POST_request.png	The single harassment POST packet (83601)
11	10a.POST_full.png / 10b.POST_full.png	Full verbose packet detail of frame 83601 (Ethernet, IP, TCP layers)
12	11.POST_inspect.png / 11.POST_Grep.png	Decoded POST form fields, two independent extraction methods
13	TCP_stream.png	Follow TCP Stream (tcp.stream eq 1707): full request/response, User-Agent, server redirect
14	12.MAC_raw.png	Byte-level MAC verification via hex dump
15	13.MAC_packets.png	MAC activity-profile packet counts
16	14.MAC_packets_sent.png / 14.MAC_packets_sent_000.png	First-observed MAC activity (ARP, DNS, TCP handshake)
17	15.MAC_SOURCE_destination.png	Source/destination MAC fields for the POST packet
18	16.HTTP_cookie.png	Available cookie fields in this tshark build
19	17.cookies.png	Raw "contains @" filter: 149 packets, including false positives
20	17.cookies_capture.png	Representative gmailchat cookie values

#	Screenshot	Evidence Shown
21	18.email_cookies.png	Regex-filtered result: one genuine email, 109 occurrences
22	19.gmail_cookie.png	Packet 78967 in detail with the gmailchat cookie pair
23	20.Tshark_timeline.png	Composite timeline of the Gmail session and the willselfdestruct.com interaction

Appendix C — Command Reference

```
sha256sum nitroba_original.pcap | tee ../notes/hash_original.txt
md5sum nitroba_original.pcap | tee -a ../notes/hash_original.txt
cp -v nitroba_original.pcap ../working/nitroba_working.pcap
sha256sum ../working/nitroba_working.pcap
chmod 444 nitroba_original.pcap

tshark -r nitroba_working.pcap -Y 'http.request.method == "GET" && http.host == "www.willselfdestruct.com"' \
 -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e http.request.uri
tshark -r nitroba_working.pcap -Y 'data-text-lines contains "secure anonymous E-mail"'

tshark -r nitroba_working.pcap -Y 'ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"'
tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -T fields -e urlencoded-form.key -e urlencoded-form.value
tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -V | grep -i "form item"
# Wireshark: right-click packet 83601 → Follow → TCP Stream (stream 1707)

tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -T fields -e eth.src -e eth.dst
tshark -r nitroba_working.pcap -Y 'frame.number == 83601' -x
tshark -r nitroba_working.pcap -Y 'eth.addr == 00:17:f2:e2:c0:ce' | wc -l
tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce' | wc -l
tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce' | head -n 5

tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"' | wc -l
tshark -r nitroba_working.pcap -Y 'eth.src == 00:17:f2:e2:c0:ce' -T fields -e http.cookie_pair \
 | grep -oE "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}" | sort | uniq -c | sort -rn
tshark -r nitroba_working.pcap -Y 'frame.number == 78967' -V | grep -A2 -i "gmailchat"

# Timeline: field extraction per event, combined into a single chronological table (Section 6)
```

Appendix D — Chemistry 109 Roster

Student
Amy Smith
Burt Greedom
Tuck Gorge
Ava Book
Johnny Coach
Jeremy Ledvkin
Nancy Colburne
Tamara Perkins
Esther Pringle
Asar Misrad
Jenny Kant